Vivek Vardhan Saraswathi

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Summary

Aspiring AI & Software Engineer with strong proficiency in Python, Machine Learning, REST API development, and Database Management. Experienced in developing AI-driven applications including Recommendation systems and Anomaly Detection solutions using Scikit-learn, Pandas, NumPy, and Collaborative Filtering techniques. Strong foundation in Data Structures, OOP, DBMS, and software development principles.

Education

NBKR Institute of Science and Technology, Vidyanagar **2022 – 2026**
Bachelor of Technology (B.Tech) in Artificial Intelligence & Data Science **CGPA: 8.04**

Technical Skills

- **Languages:** Python, SQL
- **ML/AI:** Machine Learning, Collaborative Filtering, Anomaly Detection, Scikit-learn
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn
- **Backend:** Flask, REST APIs, JWT Authentication, SQLAlchemy
- **Core:** Data Structures, OOP, DBMS, Problem Solving
- **Tools:** Git, GitHub, VS Code, Render, Swagger/OpenAPI

Internship

Artificial Intelligence Intern – PERSEVEX

- Developed and optimized machine learning models using Python for real-world datasets
- Performed data preprocessing, feature engineering, and model evaluation
- Improved model performance through tuning and validation techniques
- Collaborated with team members to debug, test, and refine AI workflows

Projects

CineMatch – AI-Driven Movie Recommendation System

- Developed an intelligent movie recommendation engine using User-User and Item-Item Collaborative Filtering with Cosine Similarity and Singular Value Decomposition (SVD).
- Analyzed the MovieLens dataset through rating distributions, long-tail analysis, utility matrix construction, and sparsity measurement.
- Evaluated recommendation performance using RMSE and Precision@K and implemented a Top-N recommendation engine for personalized movie suggestions.

SecurePay – AI Credit Card Fraud Detection System

- Developed an AI-driven financial fraud detection system using unsupervised anomaly detection techniques including Isolation Forest and Local Outlier Factor (LOF).
- Performed data preprocessing and feature scaling using RobustScaler to handle transaction amount outliers on highly imbalanced data.
- Evaluated anomaly detection performance using Precision, Recall, F1-Score, confusion matrices, and Precision-Recall analysis for identifying suspicious transactions.

Certifications

- **AI Fundamentals** – IBM | Learned machine learning workflows, neural networks, and AI fundamentals.
- **Gen-AI** – Oracle | Acquired knowledge of LLMs, prompt engineering, and generative AI applications.
- **Data Science** – Oracle | Specialized in data analysis, statistical modeling, and data-driven problem solving.
- **Critical Thinking & Problem Solving** – edX | Developed structured analytical thinking and decision-making techniques.